EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 07

Basic Maths and Calculus

Graph Shapes and Physics Applications

By - Excellence Programme

11 September 2026

Topics to be covered

01 Spring-force graph and perpendicular lines

02 Modulus graphs and transformations

03 Rectangular hyperbola and inverse relations

04 Parabola, square-root graph and transformations

05 Graphs used in mechanics, electricity and thermal physics

06 Circle and S.H.M. homework with guided answers

Only the graph tools we need

rising line: positive slope falling line: negative slope

horizontal line: slope 0 vertical line: slope undefined

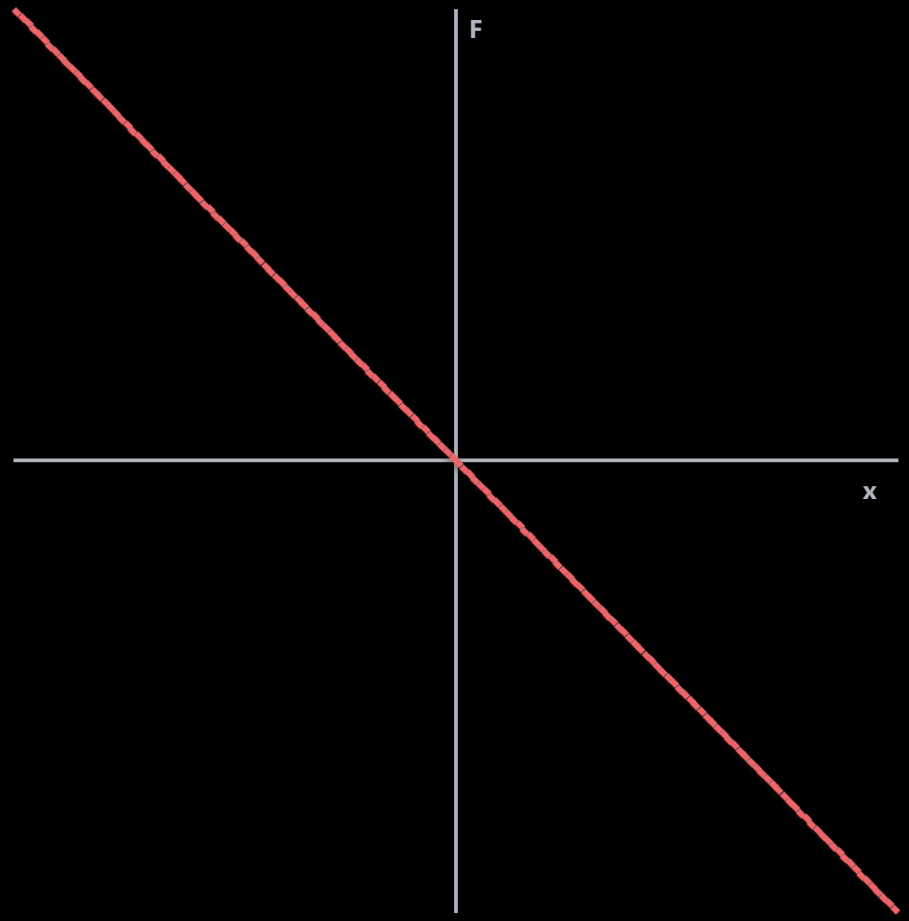
To draw any graph: make a small value table, mark points, join smoothly.

Spring force: F = -kx

Hooke's law says the spring force is opposite to displacement.

F = -kx

The graph is a straight line through the origin with negative slope -k.



Theory of the spring-force graph

1. x is displacement from the equilibrium position.

2. k is the spring constant. A larger k means a stiffer spring.

3. The minus sign means force acts opposite to displacement.

x>0 gives F<0 and x<0 gives F>0

Slope of the F-x graph = -k. Its magnitude gives the spring constant.

Rough work / extra work

Perpendicular straight lines

Two perpendicular lines meet at 90 degrees.

$$m1 \times m2 = -1$$

If m1 = tan(theta), then m2 = tan(theta+90 degrees) = -1/tan(theta)

This rule is used only when both slopes are defined.

Theory of perpendicular slopes

Rotating a line by 90 degrees changes its direction completely.

m1=tan(theta)

m2=tan(theta+90 degrees)=-cot(theta)=-1/tan(theta)

Therefore m1 x m2 = tan(theta) x [-1/tan(theta)] = -1

A horizontal line is perpendicular to a vertical line; the product rule cannot be used because vertical slope is undefined.

Perpendicular-line check

A line has slope 2. What is the slope of a perpendicular line?

Simple solution

Step 1: Use m1 x m2 = -1

Step 2: 2 x m2 = -1

Step 3: m2 = -1/2

Perpendicular slope = -1/2

Meaning of modulus

Modulus gives distance from zero, so it can never be negative.

$|x| = x$ for $x \geq 0$ and $|x| = -x$ for $x < 0$

x	-2	-1	0	1	2
$\|x\|$	2	1	0	1	2

Rough work / extra work

Theory of modulus graphs

Modulus changes every negative output into its positive magnitude.

For x>=0: |x|=x, so use the line y=x.

For x<0: |x|=-x, so use the line y=-x.

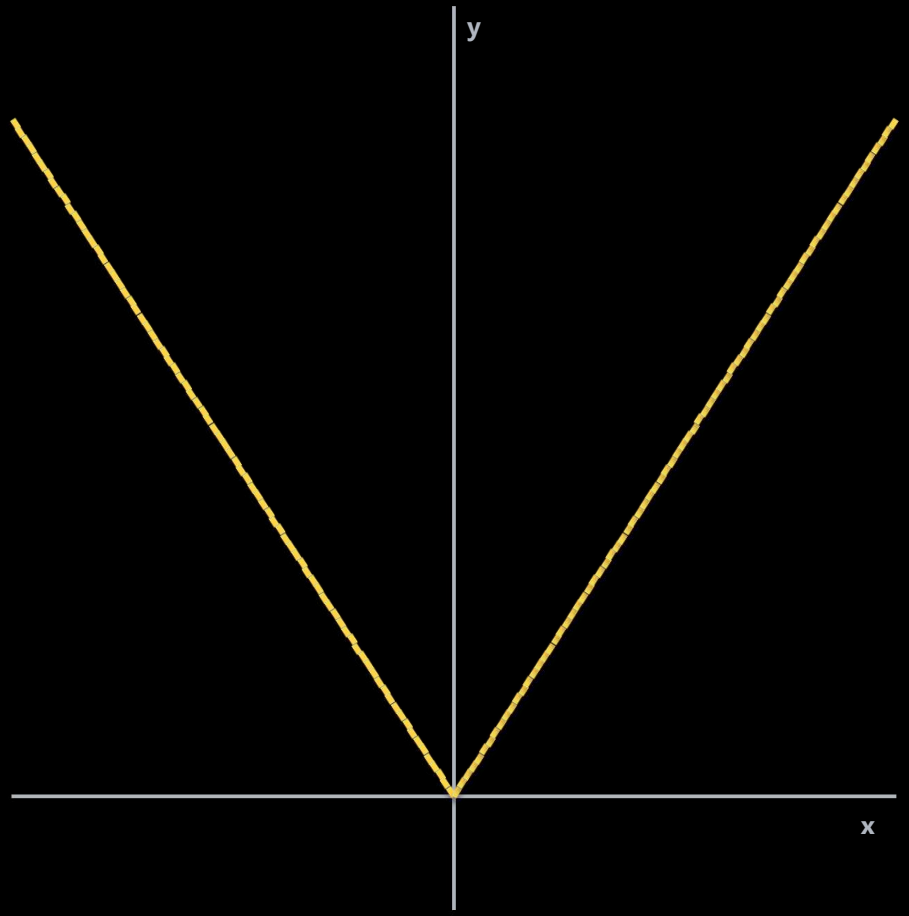
The two straight parts meet where the expression inside the modulus becomes zero.

For y=a|x-h|+k: vertex=(h,k); sign of a gives up/down and |a| gives steepness.

Graph of y = |x|

The graph is V-shaped, has vertex (0,0), and lies on or above the x-axis.

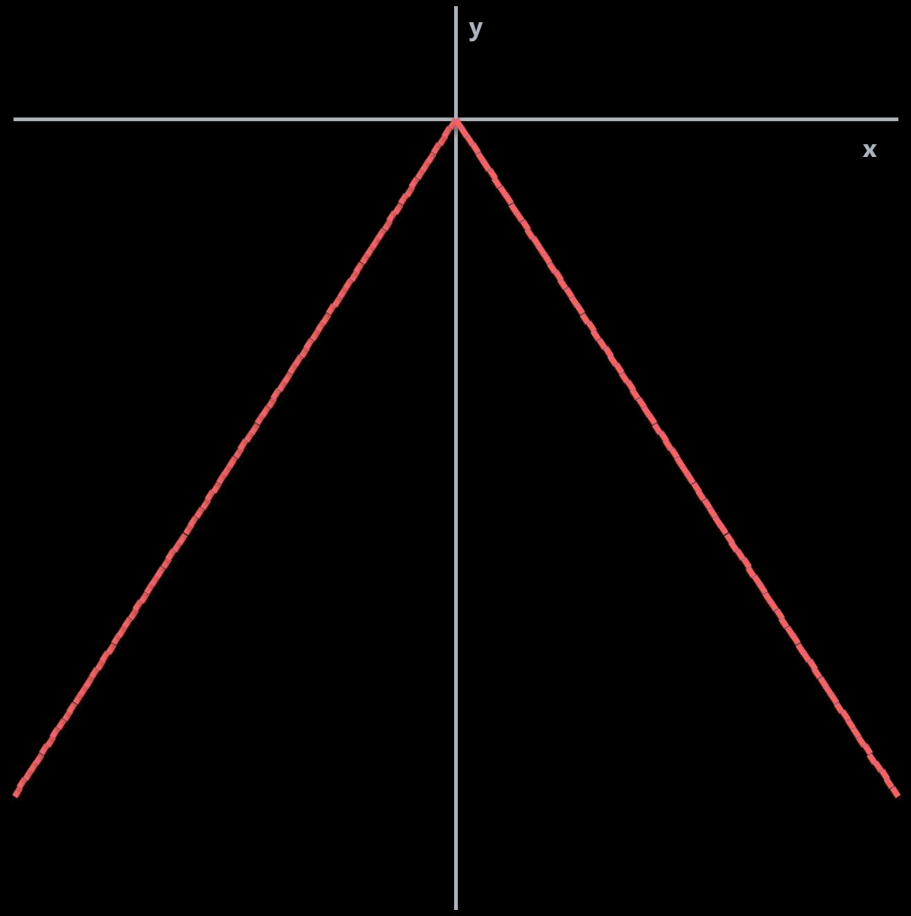
left branch: y=-x **right branch: y=x**



Graph of y = -|x|

The minus sign reflects y=|x| in the x-axis.

vertex stays at (0,0); the V opens downward



Draw y = |x-2|

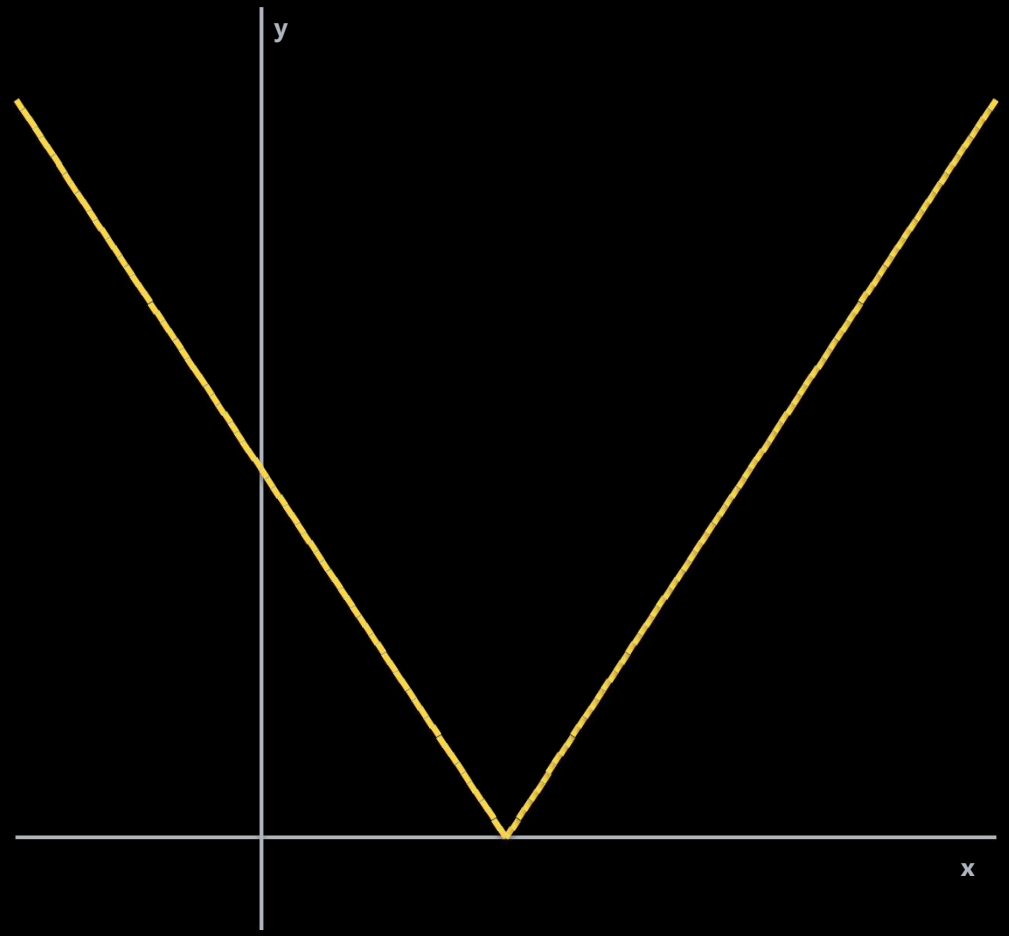
Make a value table and draw y=|x-2|.

Simple solution

x	0	1	2	3	4
y	2	1	0	1	2

Step 1: The smallest value is 0 at x=2.

Shifted 2 units right; vertex=(2,0).



Rough work / extra work

Draw y = |4-2x|

Find the vertex and draw y=|4-2x|.

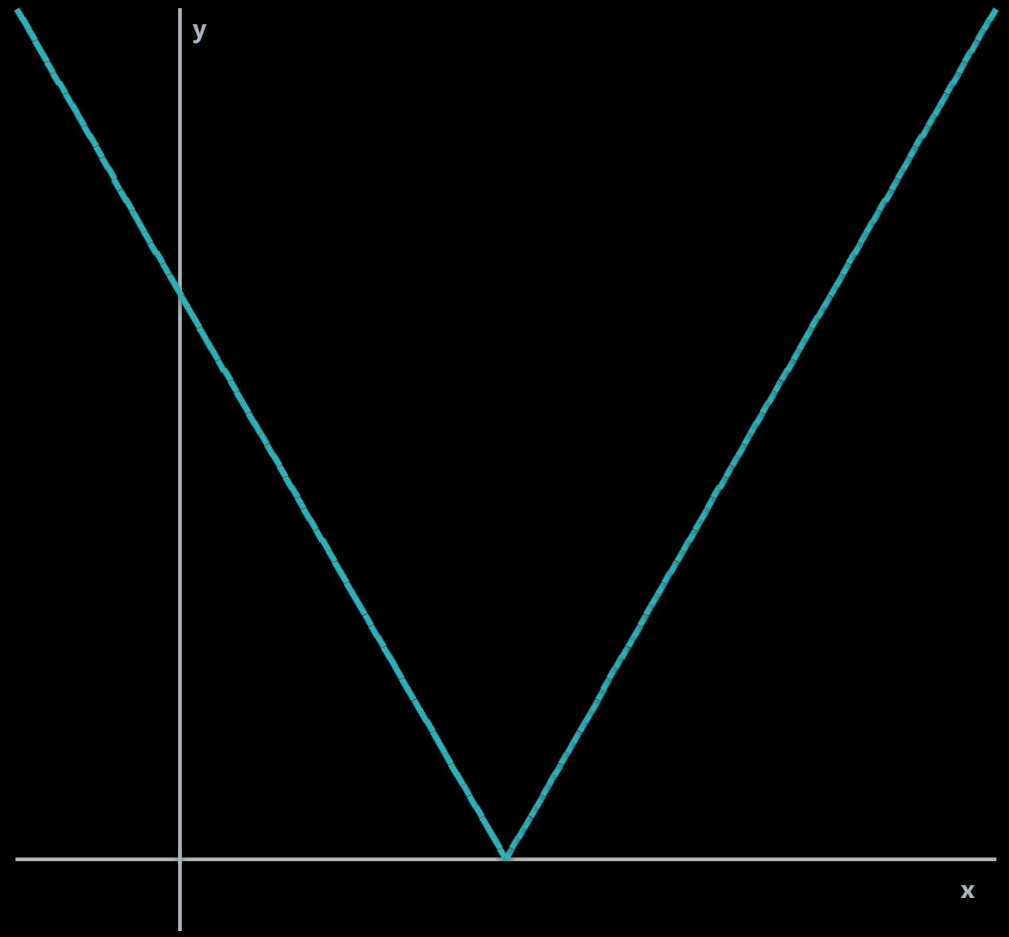
Simple solution

Step 1: Set the inside equal to zero: 4-2x=0

Step 2: x=2, so the vertex is (2,0)

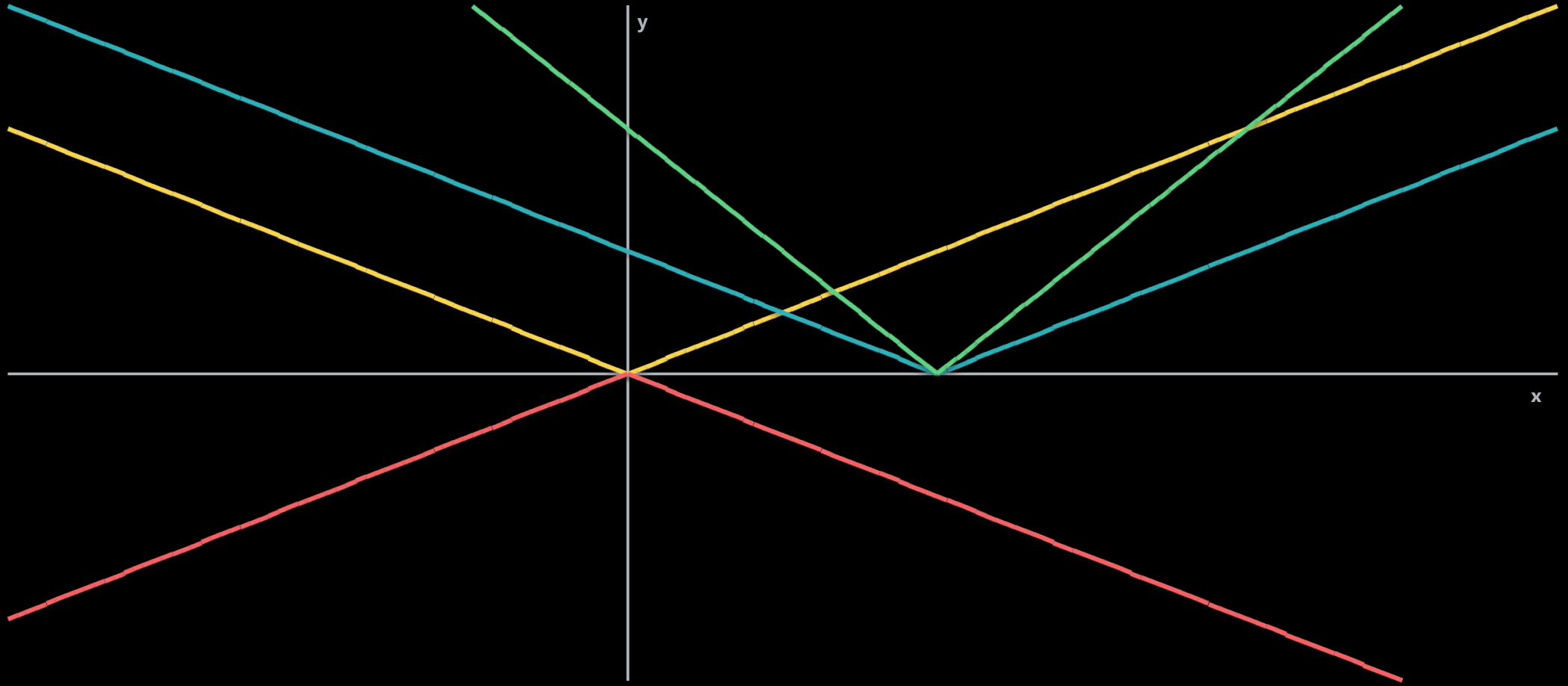
Step 3: y=|4-2x|=2|x-2|

V-shape, vertex (2,0), steeper.



Position and steepness of modulus graphs

Gold: |x| Red: -|x| Teal: |x-2| Green: 2|x-2|

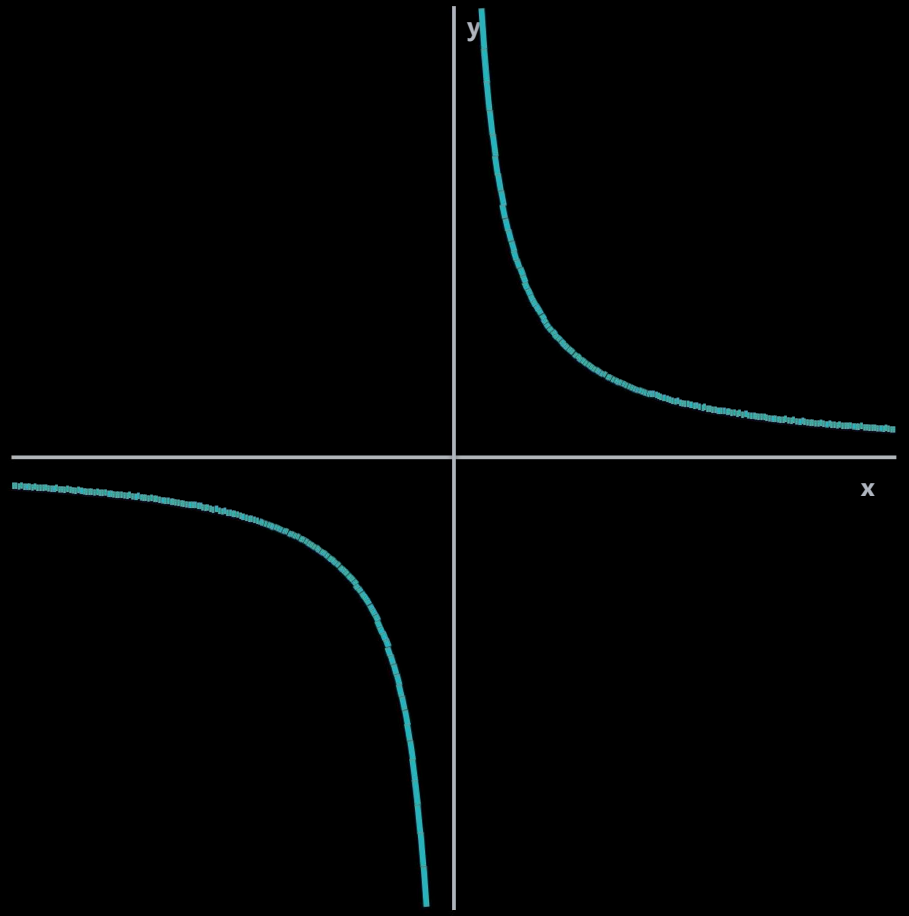


Rectangular hyperbola: y = k/x

In an inverse relation, one quantity increases while the other decreases.

y = k/x or xy = k

The curve approaches both axes but never touches them.



How to draw an inverse graph

1. Write the equation in the form y=k/x.

2. Make a value table. Avoid x=0 because division by zero is undefined.

3. Mark positive and negative x values.

4. Join smoothly; do not let the curve touch an axis.

Rough work / extra work

Inverse-square relation

A force follows F=k/r^2. What happens when r becomes 2r?

Simple solution

Step 1: Original force: F=k/r^2

Step 2: New force: F'=k/(2r)^2=k/(4r^2)

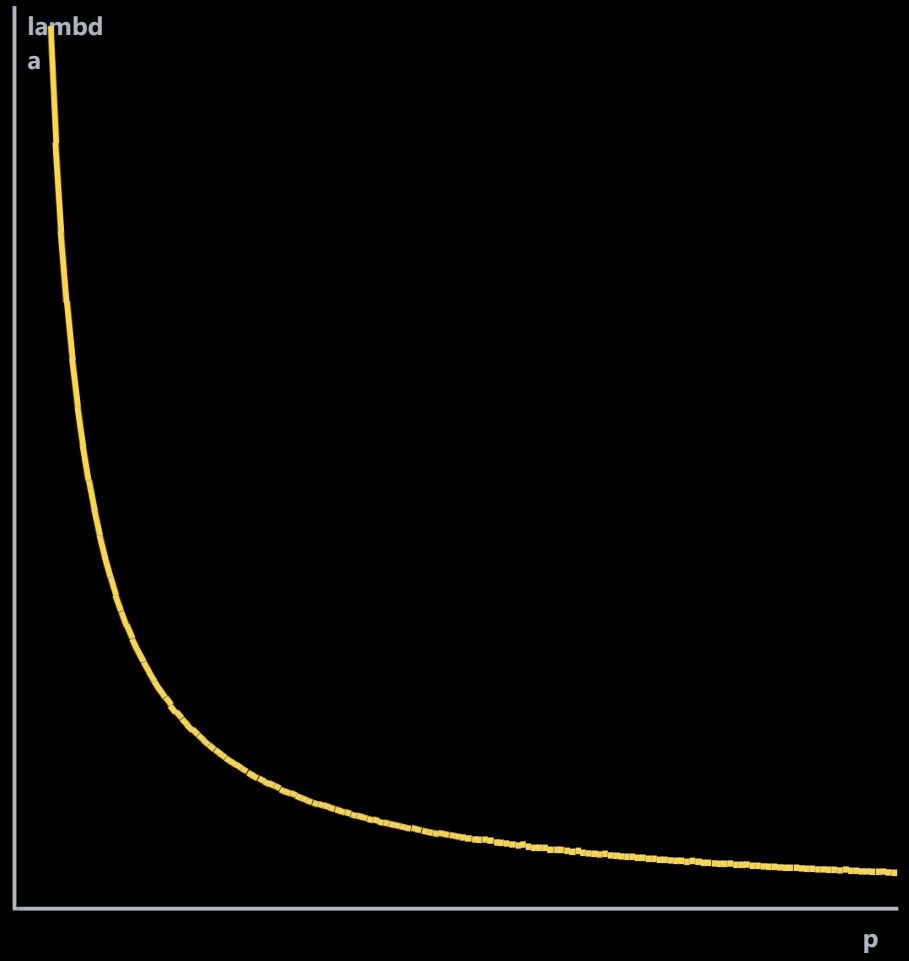
F'=F/4. Doubling distance makes force one-fourth.

De Broglie wavelength

A moving particle has wavelength inversely proportional to momentum

lambda = h/p

So lambda versus p is a decreasing rectangular hyperbola.



Wavelength and kinetic energy

K=p^2/(2m) gives p=sqrt(2mK)

lambda=h/p=h/sqrt(2mK)

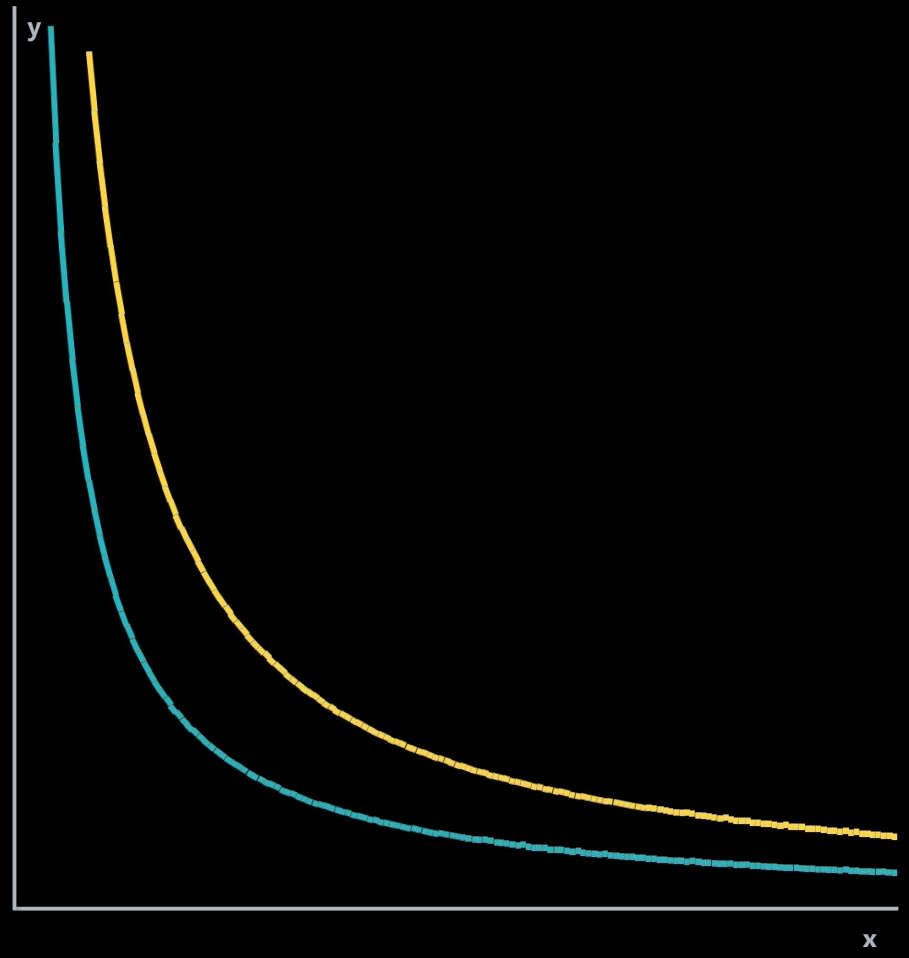
lambda is proportional to 1/sqrt(K)

Therefore lambda decreases when kinetic energy increases.

y = k/x for different k

Every positive k gives a similar decreasing curve.

A larger k places the curve farther from the origin.



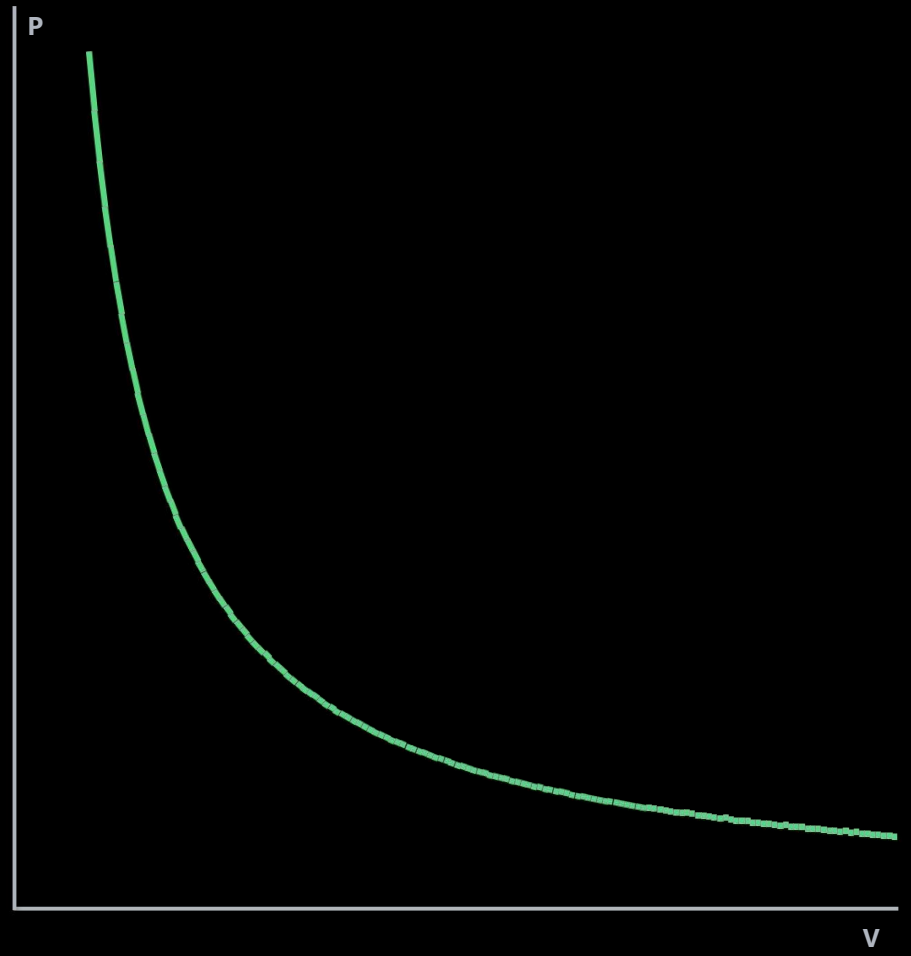
Rough work / extra work

Pressure-volume isotherm

For a fixed amount of ideal gas at constant temperature:

PV=nRT gives P=nRT/V

P is inversely proportional to V, so the graph is a rectangular hyperbola.



Theory of an isothermal P-V graph

The amount of gas n and temperature T stay constant, so nRT is one constant.

PV=constant therefore P=constant/V

If volume doubles, pressure becomes half. If volume triples, pressure becomes one-third.

Pressure becomes very large near V=0 and approaches zero for very large V.

A higher-temperature isotherm lies farther from the origin because nRT is larger.

Two gas isotherms

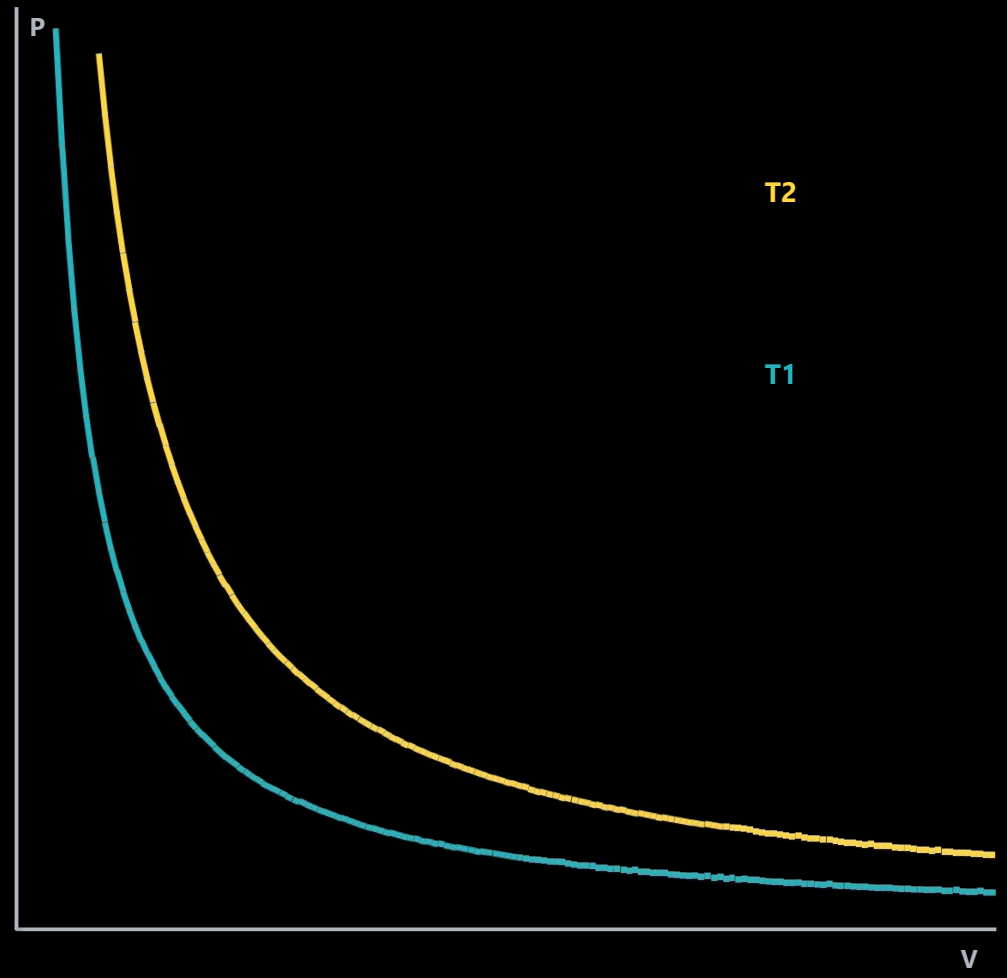
Two P-V curves are drawn for temperatures T1 and T2. The T2 curve is farther from the origin. Which temperature is higher?

Simple solution

Step 1: At the same V: P=nRT/V

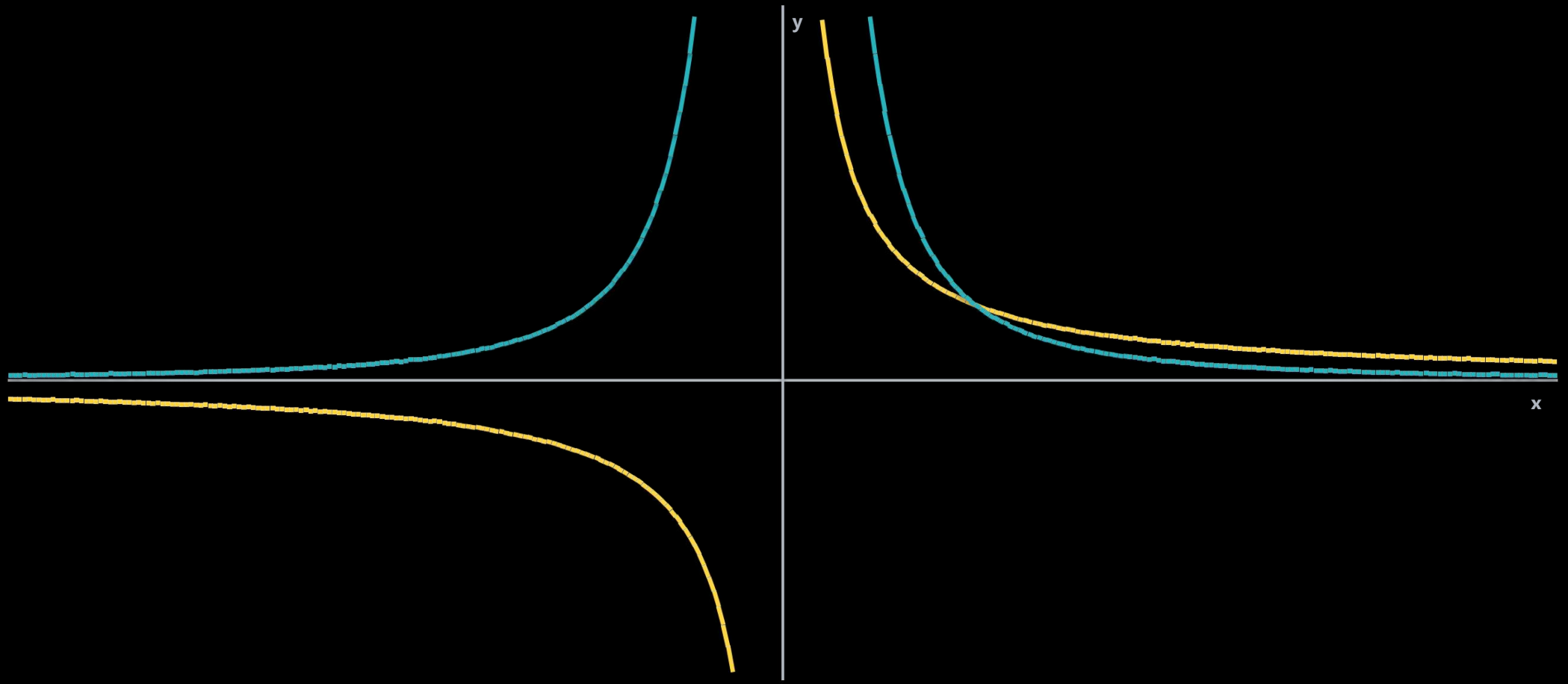
Step 2: Higher T gives higher P.

Farther curve: T2>T1.



y=1/x and y=1/x^2

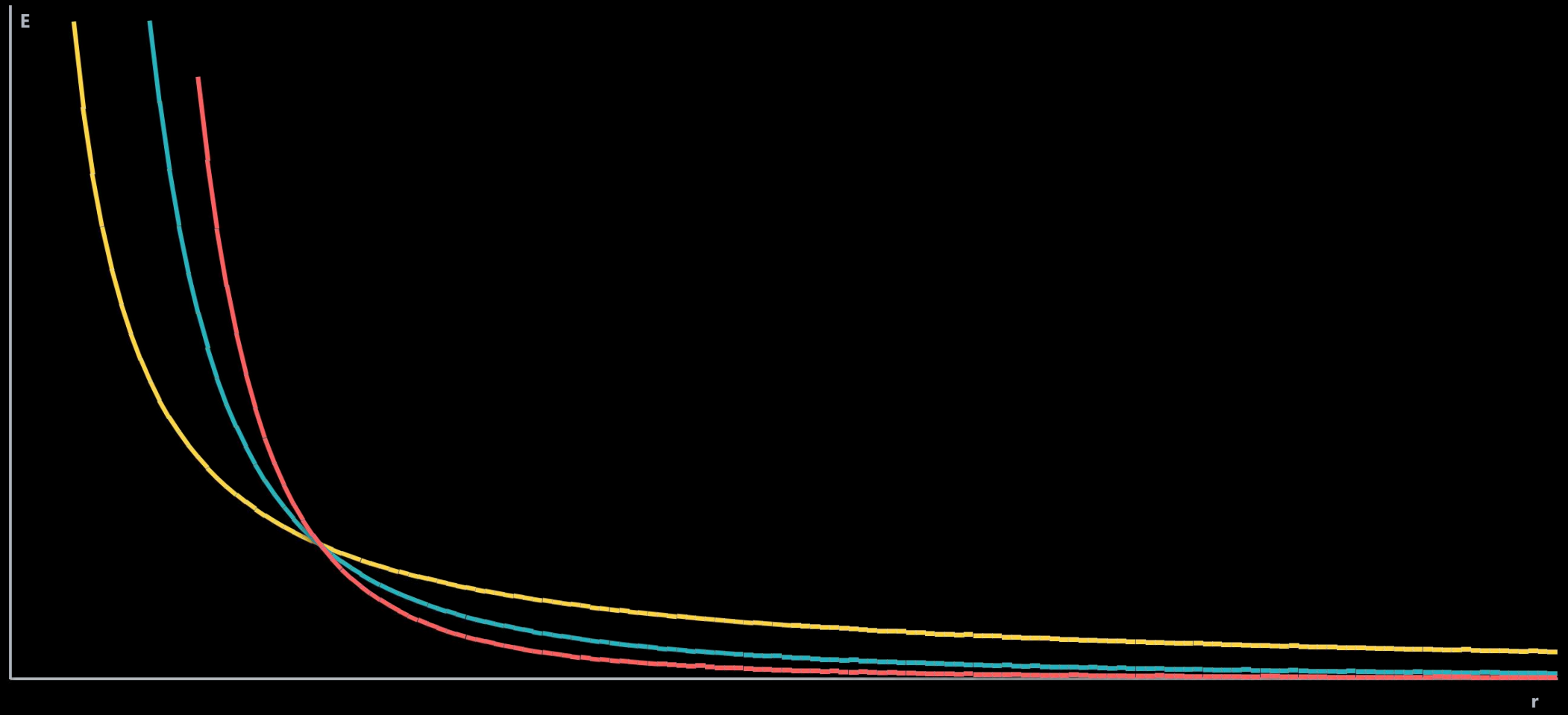
Gold: y=1/x Teal: y=1/x^2. Both approach the axes without touching them.



Rough work / extra work

Electric field versus distance

Gold: 1/r Teal: 1/r^2 Red: 1/r^3



Theory of electric-field falloff

As distance increases, the same electric effect spreads over a larger region.

line charge: E is proportional to 1/r

point charge: E is proportional to 1/r^2

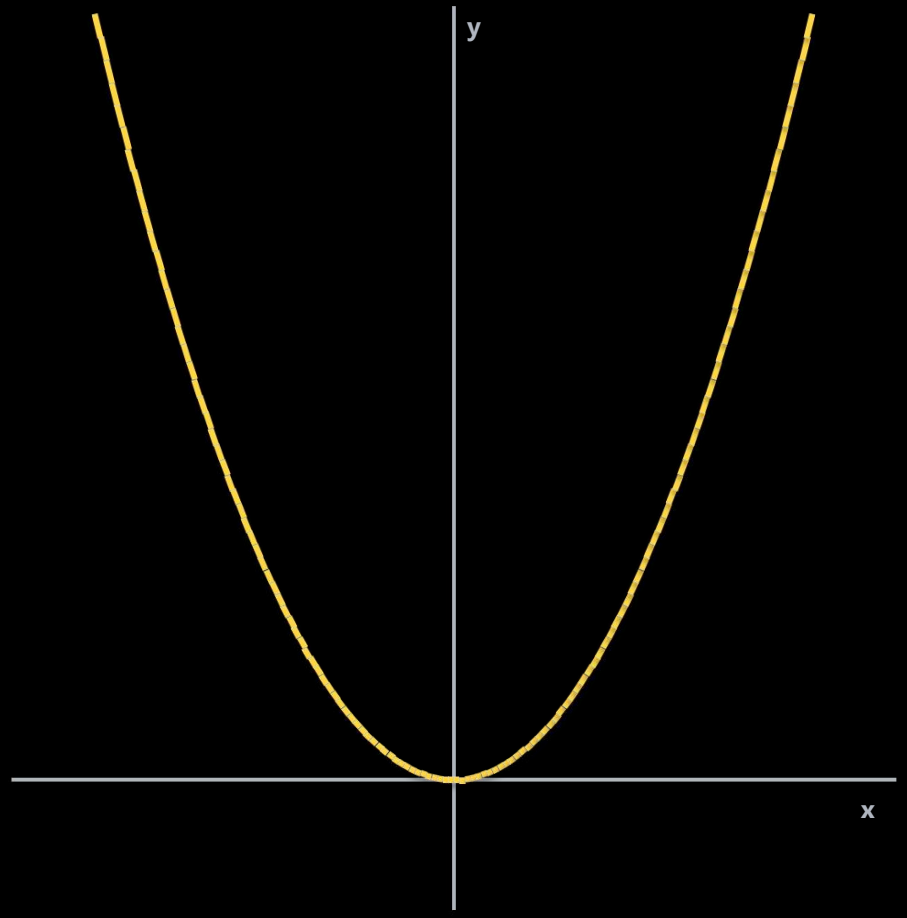
dipole far away: E is proportional to 1/r^3

At large r, 1/r^3 decreases fastest and 1/r decreases slowest.

Parabola: y = x^2

Squaring gives the same y-value for +x and -x.

x	-2	-1	0	1	2
y=x^2	4	1	0	1	4



Theory of a parabola

The general vertical parabola is y=a(x-h)^2+k.

vertex=(h,k) **axis of symmetry: x=h**

a>0: opens upward **a<0: opens downward**

A larger |a| makes the parabola narrower; a smaller |a| makes it wider.

Find the vertex first, then calculate symmetric points on both sides.

Draw y = x^2 - 4

Find the vertex and x-intercepts of y=x^2-4, then sketch the graph.

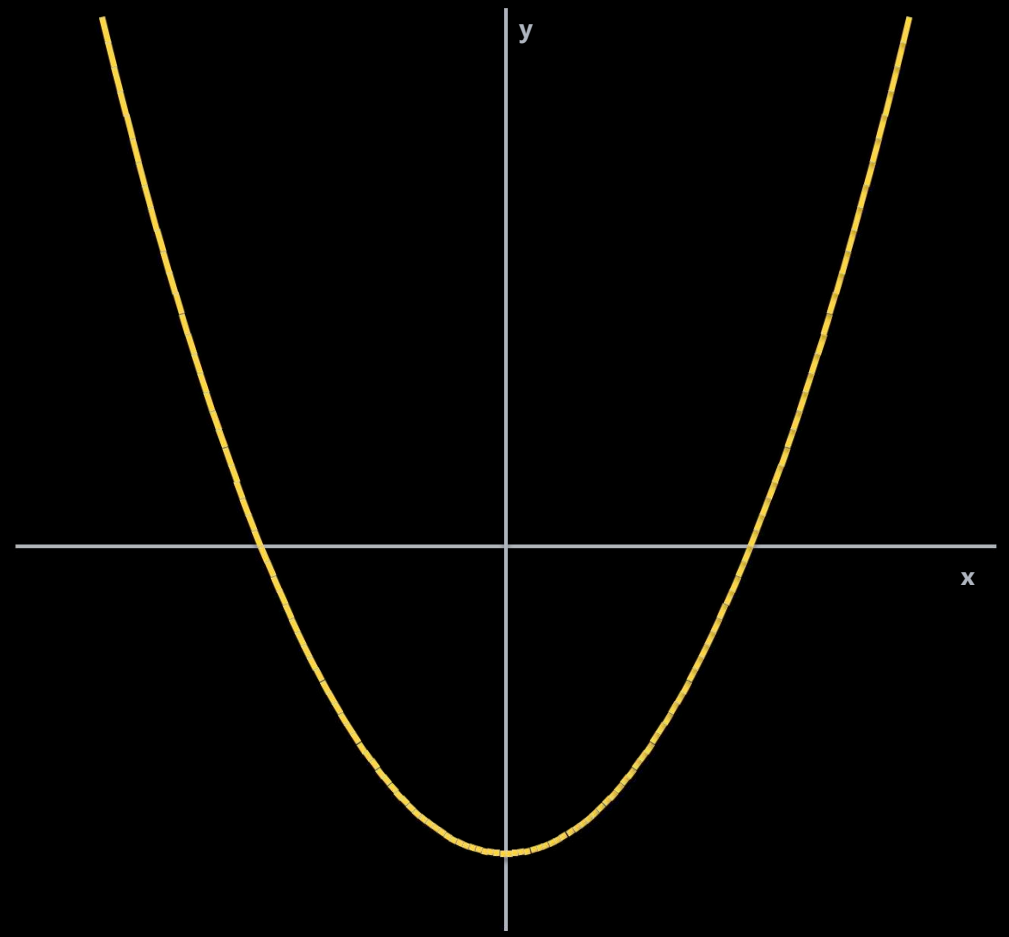
Simple solution

Step 1: Vertex: x=0 gives y=-4, so vertex=(0,-4)

Step 2: For x-intercepts: x^2-4=0

Step 3: x=+2 or x=-2

Cuts x-axis at (-2,0) and (2,0).



Rough work / extra work

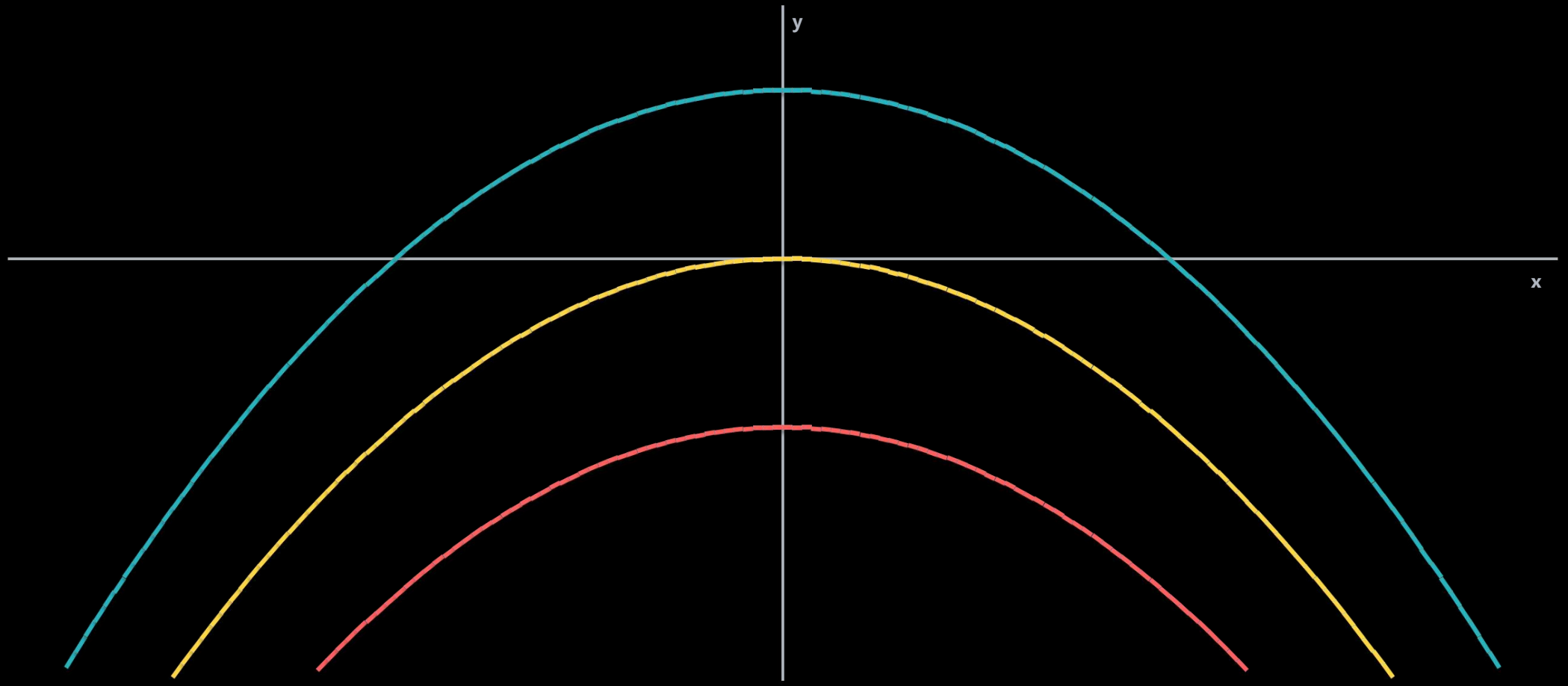
Vertical shifts of y=x^2

Gold: y=x^2 Teal: y=x^2+4 Red: y=x^2-4



Downward parabolas

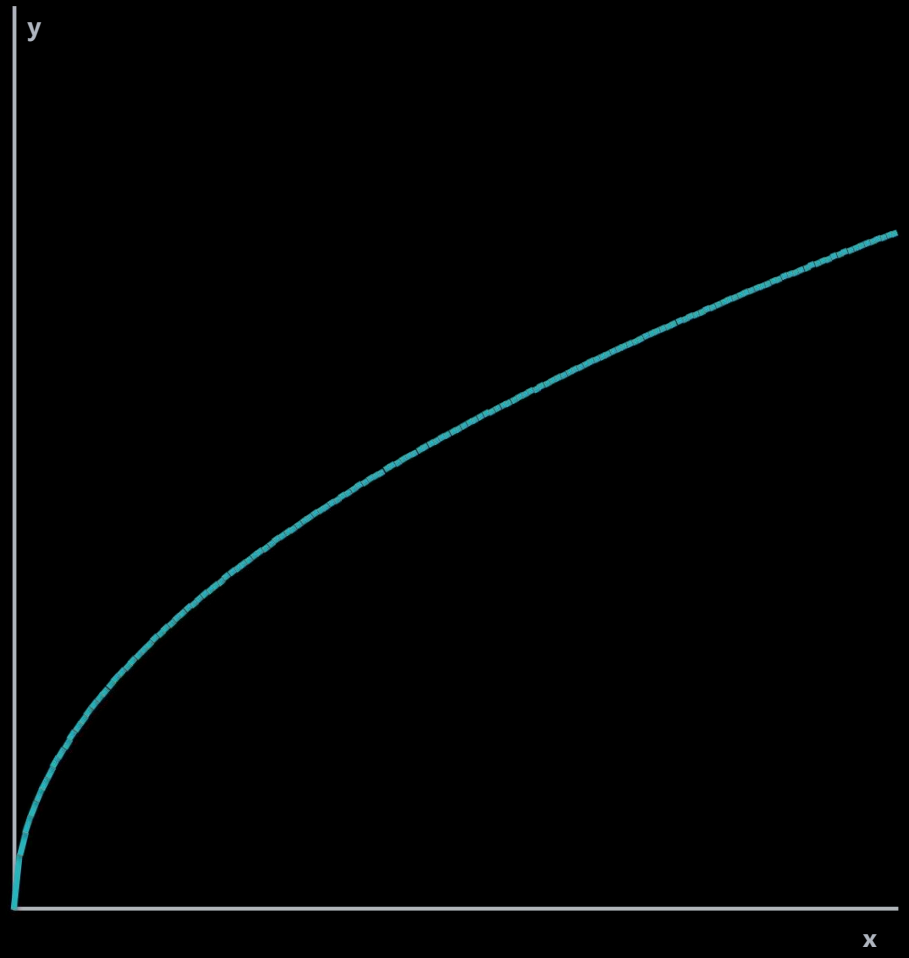
Gold: -x^2 Teal: -x^2+4 Red: -x^2-4



Square-root graph: y = sqrt(x)

A real square root needs x>=0.

x	0	1	4	9
sqrt(x)	0	1	2	3



Theory of the square-root graph

y=sqrt(x) is the inverse of y=x^2 when x is restricted to x>=0.

domain: x>=0 range: y>=0

It begins at (0,0), rises quickly, and then becomes flatter.

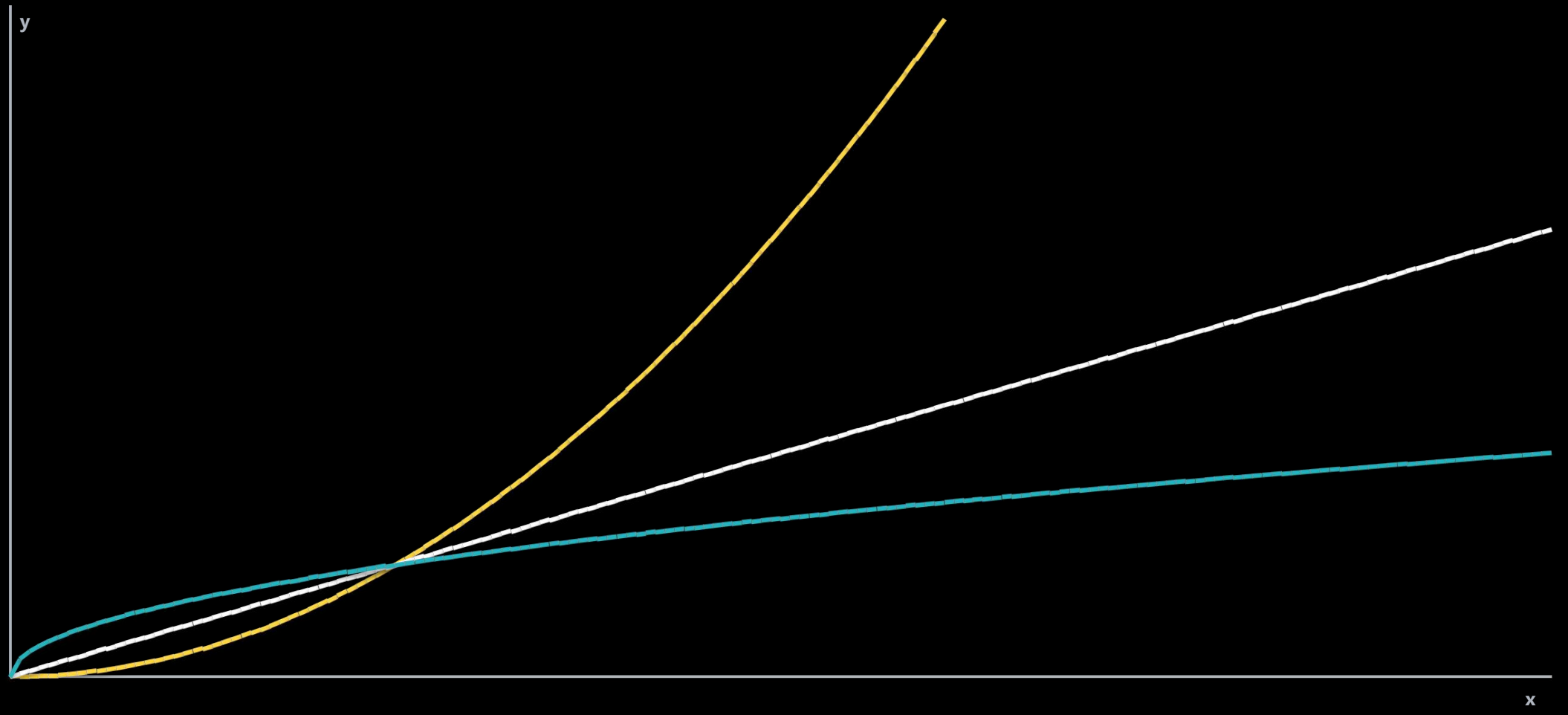
y=sqrt(x-h)+k starts at (h,k)

For real-number graphs, the expression inside the square root cannot be negative.

Rough work / extra work

y=x^2, y=x and y=sqrt(x)

Gold: y=x^2 White: y=x Teal: y=sqrt(x). All meet at (0,0) and (1,1).

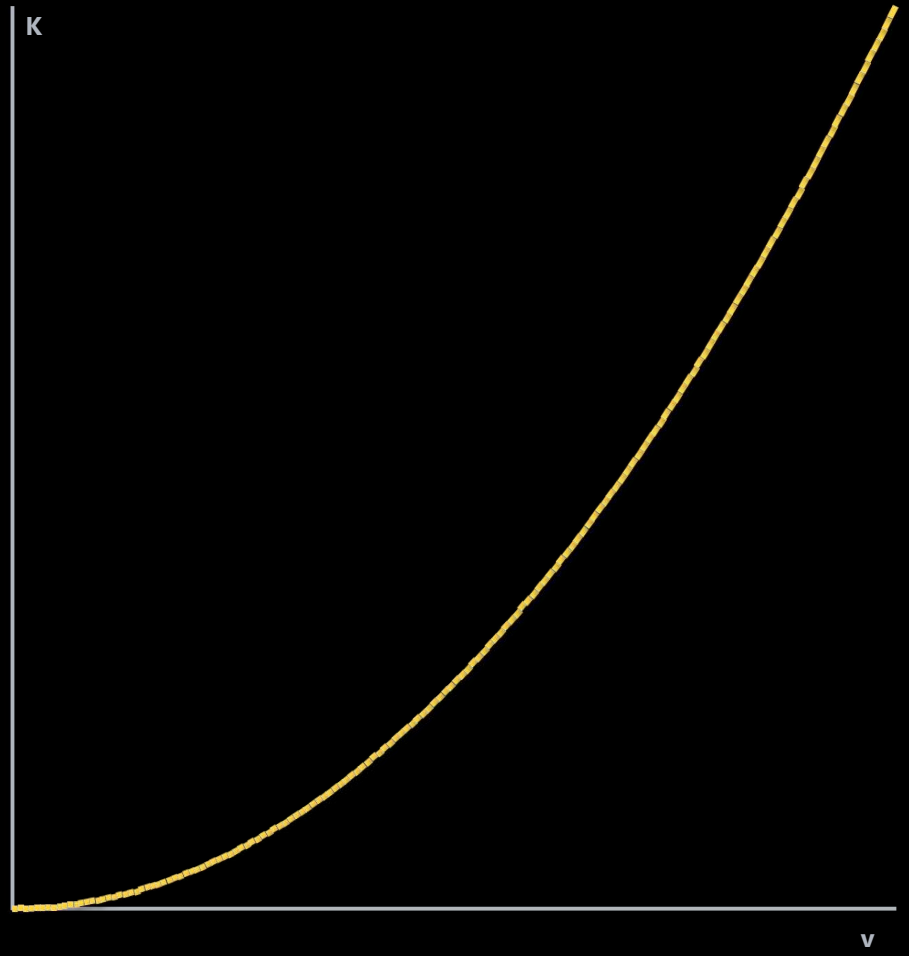


Kinetic energy versus speed

For constant mass:

$$K = (1/2)mv^2$$

K is proportional to v^2, so K versus v is an upward parabola through the origin.



Theory of the kinetic-energy graph

Mass is constant, so K depends on the square of speed.

v becomes 2v -> K becomes 4K

v becomes 3v -> K becomes 9K

Speed cannot be negative, so the physical K-v graph uses only v>=0.

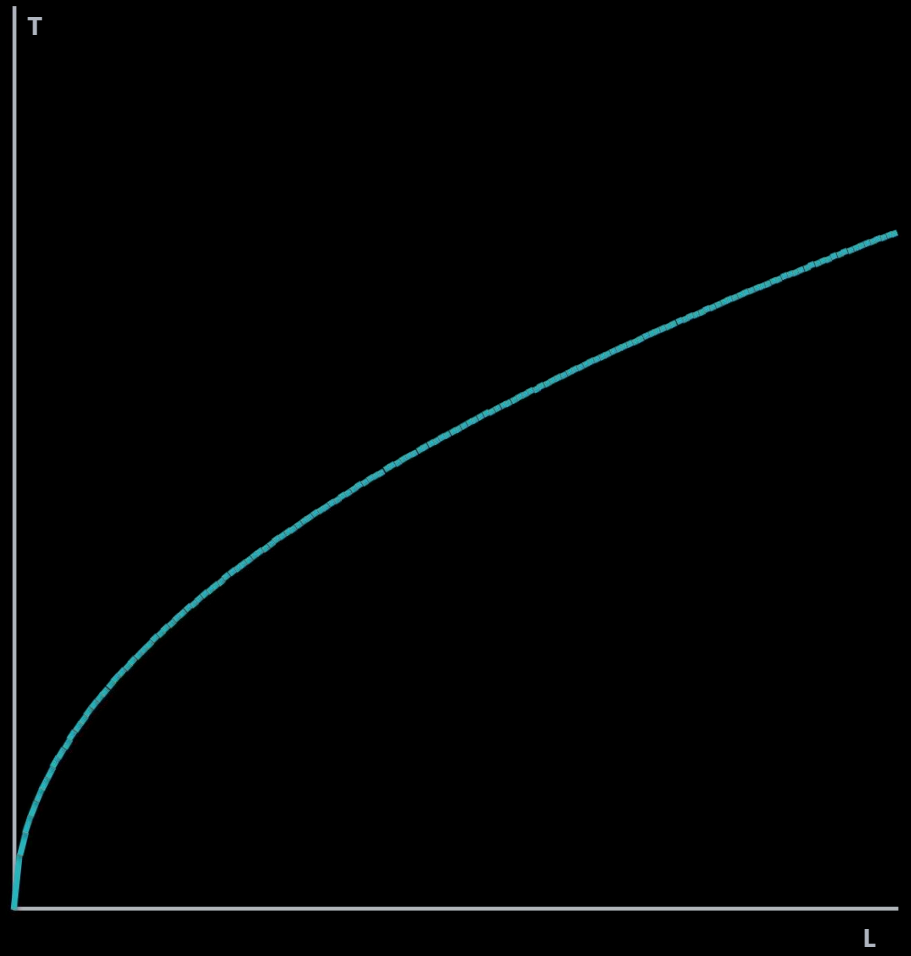
The curve starts at the origin and becomes steeper as speed increases.

Simple pendulum: T versus length

For small oscillations:

$$T = 2\pi \sqrt{L/g}$$

T is proportional to sqrt(L): it rises, but its slope gradually decreases.



Theory of the pendulum graph

For one place, g is constant. The period depends on the square root of length.

T is proportional to sqrt(L)

L becomes 4L -> T becomes 2T

L becomes 9L -> T becomes 3T

T^2 versus L is a straight line through the origin.

Rough work / extra work

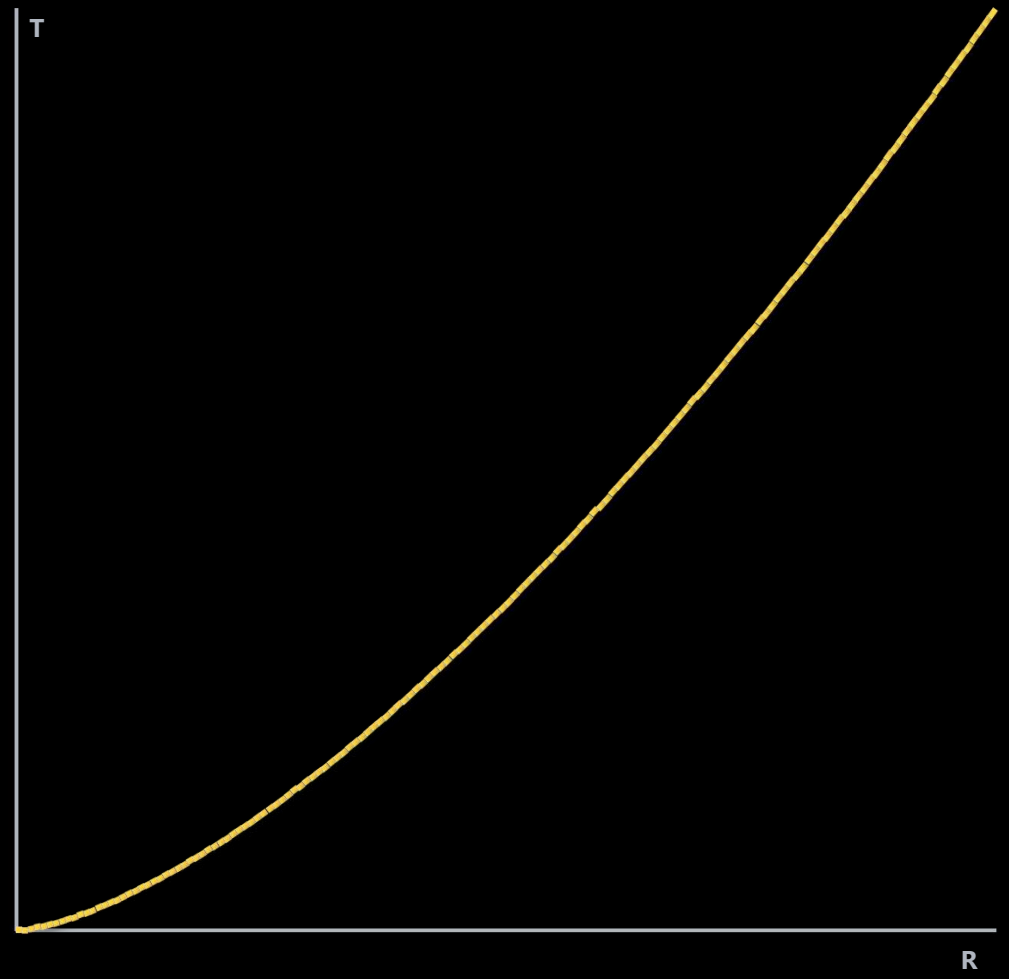
Kepler's third law

For planets orbiting the same central body:

T^2 is proportional to R^3

T is proportional to R^(3/2)

T versus R is a rising curved graph.



Theory of Kepler's third-law graph

T is orbital period and R is the orbital radius or semi-major axis.

T^2/R^3 = constant for bodies orbiting the same central mass

Therefore T^2 plotted against R^3 gives a straight line through the origin.

But T plotted directly against R is a curved graph because T is proportional to R^(3/2).

The constant changes if the central mass changes.

Kinetic energy versus mass

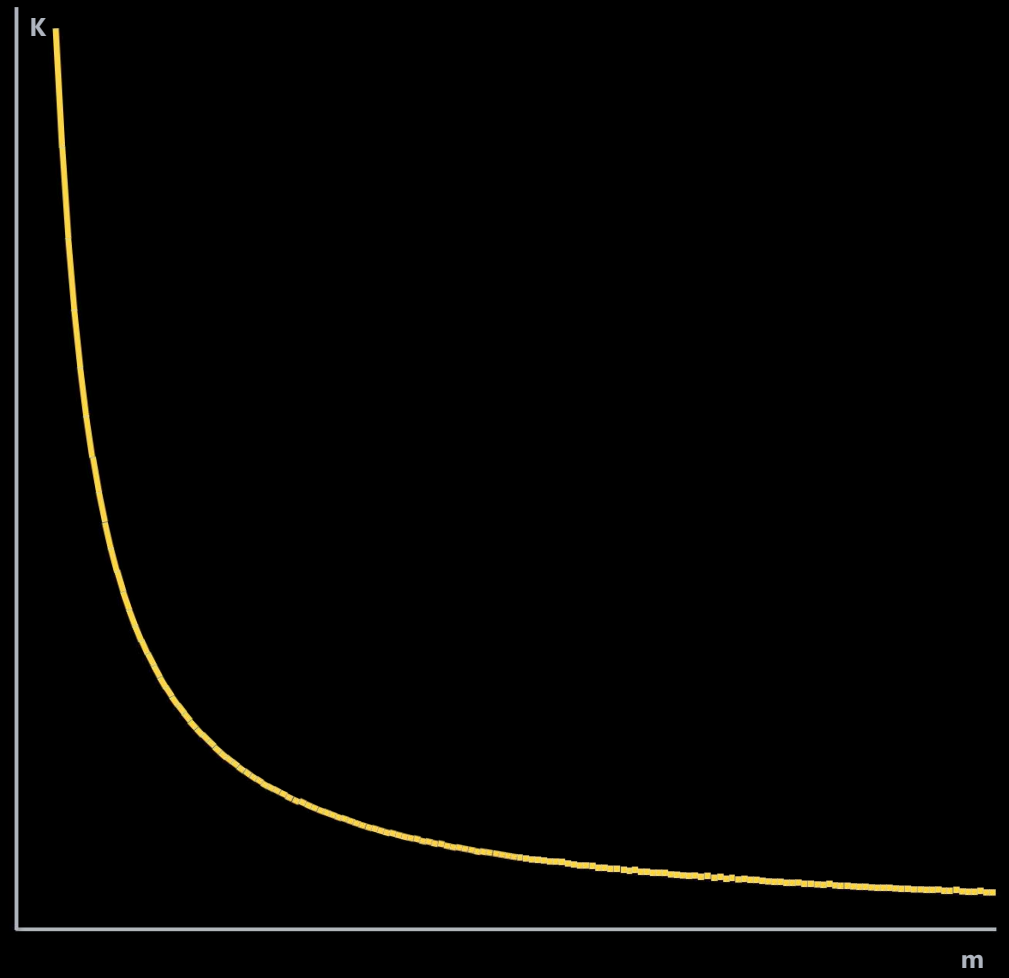
Momentum p is constant. Draw the relation between kinetic energy K and mass m.

Simple solution

Step 1: K=p^2/(2m)

Step 2: With p constant, K is proportional to 1/m.

Decreasing rectangular hyperbola.



Homework: draw x^2 + y^2 = 9

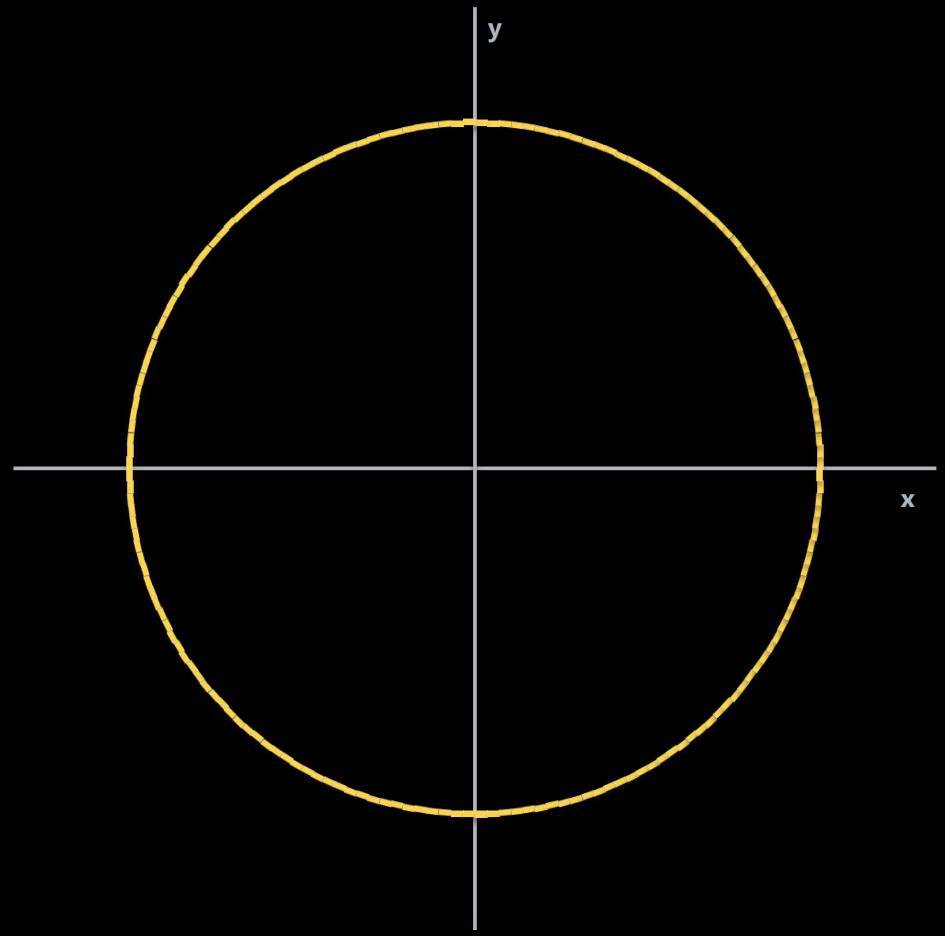
Identify the curve, its centre and its radius.

Simple solution

Step 1: Compare with x^2+y^2=R^2

Step 2: R^2=9, so R=3

Centre (0,0), radius 3.



Rough work / extra work

Homework: S.H.M. velocity-position graph

For v=omega sqrt(A^2-x^2), draw the graph between v and x.

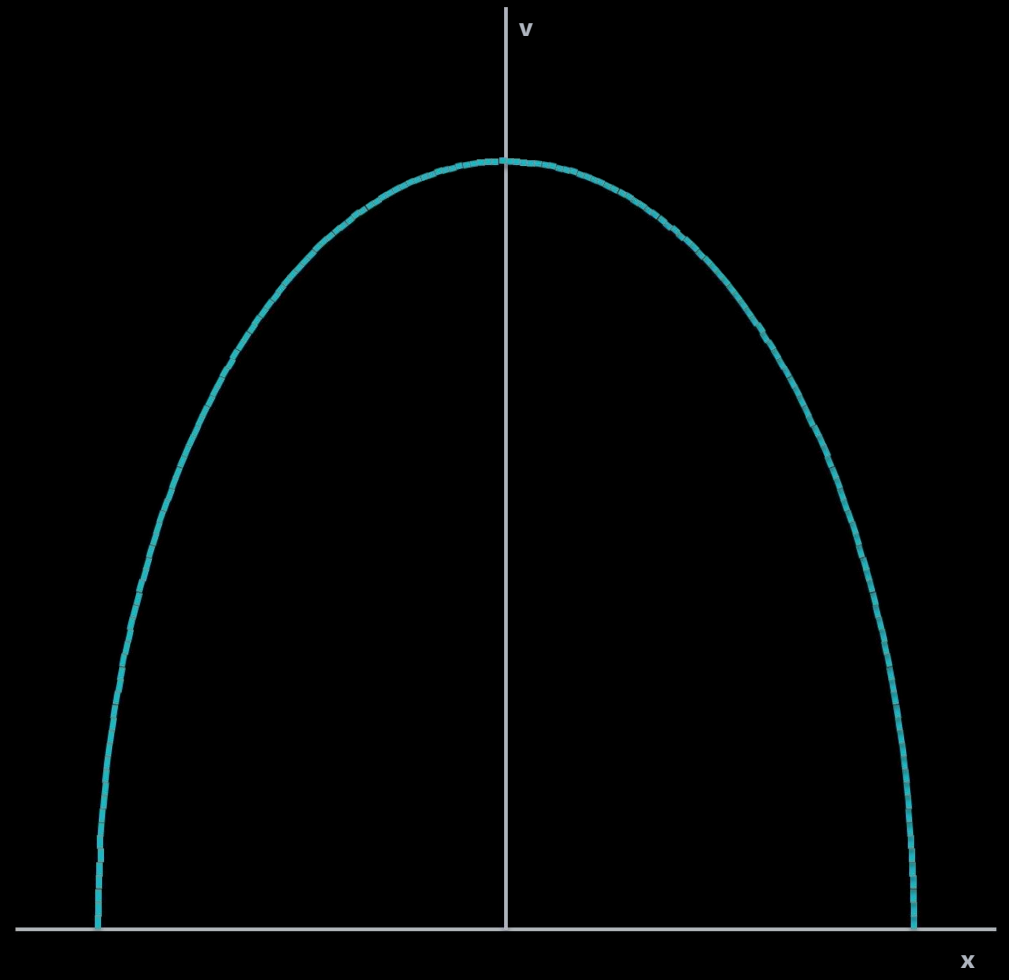
Simple solution

Step 1: v^2=omega^2(A^2-x^2)

Step 2: x^2/A^2 + v^2/(omega A)^2 = 1

Upper half of an ellipse because speed is non-negative.

Maximum at x=0; zero at x=+A and -A.



Graph-shape lock lines

|x| -> V shape 1/x -> rectangular hyperbola

x^2 -> parabola sqrt(x) -> rising curve for x>=0

inverse power: larger denominator power -> faster decrease

m1 x m2 = -1 for perpendicular non-vertical lines

Practice

Draw every graph once using a small value table.

Label both axes and mark important points or intercepts.

Complete the circle and S.H.M. questions without looking at the answer slide.

Rough work / extra work